Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> First let's calculate the distance John traveled before he turned around. He drove for 3 hours at a speed of 60 mph. Using the formula distance =speed*time. distance = 60*3 = 180 miles. Now let's calculate the distance he traveled during his attempt to get home. He spent the first 2 hours in standstill traffic. During the first 2 hours in standstill traffic he traveled 0 miles. Then he spent the next half-hour driving at a speed of 30 mph. Using the formula distance =speed*time. distance = 30*0.5 = 15 miles. Now let's calculate the remaining time he spent driving. He had a total of 4 hours to get home. He already spent 2 hours in traffic and 0.5 hours driving. Therefore 4-2-0.5 = 1.5 hours. During this time he was driving at a speed of 80 mph. Using the formula distance =speed*time. distance = 80*1.5 = 120 miles. Now let's calculate the total distance he traveled. He initially traveled 180 miles. Then he traveled an additional 15 miles. And finally he traveled 120 miles. Therefore 180+15+120 = 315 miles. Now let's calculate the distance he is from home. He initially traveled 180 miles away from home. Then he traveled 15 miles back towards home. And finally he traveled 120 miles away from home. Therefore 180-15+120 = 315-15 = 300 miles. Therefore at the end of those 4 hours he is 300 miles away from home. </think><answer>300</answer>